NEBIUS × NVIDIA GLOBAL AI HACKATHON 2026 · BEST APPS & AGENTS

Verdikt.

AI reads the case. Deterministic code writes the reply. Humans decide.

AI-IN | THE LINE | DETERMINISTIC-OUT

A DECISION PIPELINE FOR REGULATED DOCUMENT WORKFLOWS

The decision is small. The *reading* is what takes days.

A patient-assistance application is a stack of four documents. Every hand-off in the chain opens all four again.

01

Enrollment form

Identity, household size, income attestation, consent.

02

Income proof

Pay stubs or a tax return, checked against the program's cap.

03

Insurance card

No coverage, or a denial for this exact drug.

04

Prescription

Prescriber, NPI, dosage and refills.

Intake, eligibility, fulfillment and compliance each re-read the same PDFs. That re-reading is the cost Verdikt removes.

Draw the line where the *trust requirement* changes.

Reading may be wrong, so it stays reviewable. Writing must be reproducible, so it never touches a model.

01 · AI-IN

AI reads

Extraction, triage, retrieval and a structured verdict. Each step has a fallback and an audit entry.

02 · DETERMINISTIC-OUT

Code writes

The reply is a merge of a frozen, human-approved template with the verdict. No model on this path.

03 · PEOPLE

Humans decide

Reviewers see every stage and approve or reject a held reply. A boundary test fails the build if the render core imports the AI layer.

READ · CLASSIFY · RETRIEVE · REASON

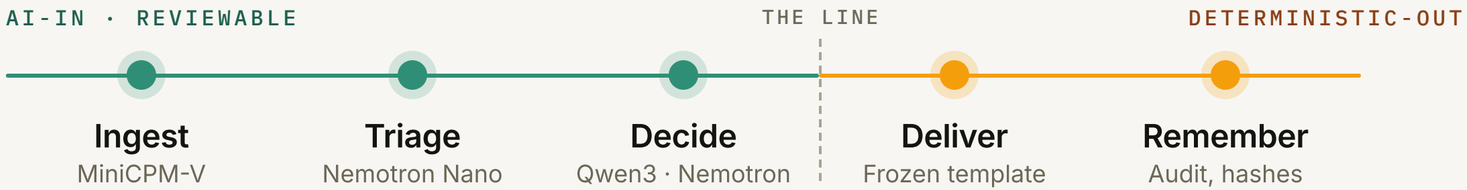
THE LINE · CITED VERDICT

MERGE INTO A FROZEN TEMPLATE

Many channels in, one case, *two artifacts* out.

IN

- **Email** via IMAP
- **JSON API**
- **MCP** tools



OUT

- **Reply letter** PDF or DOCX
- **JSON verdict** for CRM
- **Signed webhooks**

FastAPI web · background worker · Postgres · Redis · S3-compatible storage · tenant-scoped · one Nebius Token Factory endpoint. A reviewer approves or rejects a held reply in the pipeline view.

Three steps, *one endpoint*.

① READ

The models read

- **Ingest:** MiniCPM-V reads attachments into the program's field schema
- **Triage:** Nemotron Nano classifies and routes; a rule-based classifier is the fallback

② DECIDE

The rules decide

- **Retrieve:** Qwen3-Embedding finds the rules that apply
- **Rules:** Nemotron Super turns each rule into a formula the author reviews; code applies it, and the model reasons only where no formula exists
- Every verdict cites the rule that fired

③ DELIVER · REMEMBER

The template writes

- A frozen, human-approved template merged with the verdict
- No model on the render path
- One audit entry per stage, with template hash and render fingerprint

Models run on one Nebius endpoint. Rules are human-authored and versioned. Templates are frozen before they can render.

One packet in, one *sealed reply* out.

Packet arrives
Email, JSON or MCP; fax and scans as attachments

HUMAN

One case
Normalized, tenant-scoped, deduplicated

IMAP · API

Ingest reads
MiniCPM-V into the program's field schema

AI · VISION

Triage routes
Nemotron Nano, rule-based fallback

AI · LLM

Decide cites
Qwen3 retrieval, rule formulas by Nemotron Super

AI · LLM

Deliver merges
Verdict into a frozen, approved template

NO AI

Reviewer decides
Approve or reject; an audit entry per stage

HUMAN

Green steps use models and stay reviewable. The amber step is deterministic code. Dark steps are people and plumbing.

Not every case needs a *verdict*.

CASE TYPE	ROUTE	DECIDE RUNS?
New intake	Decide, then deliver	Yes
Refill	Straight to pharmacy	No
Appeal	CRM appeals queue	Yes
Missing-info reply	Decide on merged evidence	Yes
Not decidable	Reject, audit row only	No

SAME HOUSEHOLD, TWO PROGRAMS

$60,000

Cardiovita cap $86,560 (400% of the 2026 poverty guideline): **passes**.

OncoBridge cap $54,100 (250%): **fails**.

Same code, different data.

Routes are a table, overridable per program or tenant. If the triage model fails or is unsure, a rule-based classifier takes over.

Evidence, not adjectives.

934	928	0	6
tests in the suite	passing	failing	skipped

The split is enforced

Boundary tests fail the build if the render core imports the AI layer.

The labeler measurably improved

On one dense 104-field form, correct labels went from 24 to 98 with vision, chunking and crop.

Every stage is on the record

One audit entry per stage, with the model that handled it and the time it took.

Test counts from the latest full run · the labeling result is one benchmark form

Five models, *one endpoint*.

STAGE	MODEL	JOB
Ingest	openbmb/MiniCPM-V-4_5	Multimodal extraction
Ingest	NVIDIA Nemotron-3.5-Super-VL	Optional second reader; flags disagreements
Triage	NVIDIA Nemotron Nano	Case classification and routing
Decide	Qwen/Qwen3-Embedding-8B	Rule retrieval
Decide	NVIDIA Nemotron Super 120B	Rules into formulas; verdict where no formula exists

One endpoint, one key

Every model call goes through Nebius Token Factory.

Swappable by configuration

The model per stage is set per tenant or program, with no code change.

No model at render

The reply comes from a frozen template, never from a model.

Where the value comes from, and *how to measure it*.

DEMONSTRATED

- Every verdict cites the rule that fired
- Refills skip the verdict entirely
- The reply is merged, never generated
- A second program runs on the same code

BY DESIGN

- A new program is configuration, not a build
- A new channel is one adapter
- Models swap per program
- Replies are reproducible and auditable

TO MEASURE IN A PILOT

- [__]% of cases a reviewer re-opens the source PDF
- [__] median time from receipt to verdict
- [__] cost per case

The pattern generalizes: prior authorization, benefit verification, informed consent and safety reporting share the same shape, a regulated letter that is also the audit artifact.

Bracketed figures are to be filled from a measured pilot

Honest about *what is left*.

01

Harden the deploy

It is live and self-hosted, with every model call on Nebius Token Factory. Next: health checks and backups.

02

Byte-level reproducibility

Replay a stored case in CI and assert the letter's bytes, not only its structure.

03

A measured pilot

Fill the "to measure" figures from real cases.

AI reads. Deterministic code writes. Humans decide.